



University
of Exeter

COMM113

Deep Learning – Week 7

Object Detection and Image Segmentation

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Introduction to Segmentation

Semantic Segmentation: Identify groups of pixels that go together

Classification



CAT

No spatial extent

Semantic Segmentation



**GRASS, CAT,
TREE, SKY**

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation



DOG, DOG, CAT

[This image is CC0 public domain](#)

How is segmentation different from object detection?

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How is segmentation different from object detection?

Detection uses bounding boxes; segmentation assigns a label to every pixel.

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Why might bounding boxes be insufficient for tasks like autonomous driving?

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Why might bounding boxes be insufficient for tasks like autonomous driving?

Bounding boxes cannot capture road/lane boundaries.

Why do we need pixel-level understanding?

Bounding boxes are coarse

Segmentation provides spatial precision: boundaries, shapes, fine structures

Critical for:

- Self-driving
- Medical imaging
- Robotics manipulation

Introduction to Segmentation

Instance Segmentation: Identify individual objects within an image while assigning a unique label to each pixel.

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What real-world tasks need instance-level distinction?

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What real-world tasks need instance-level distinction?

E.g., tracking pedestrians, medical cell segmentation

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Application: Better Classification



E.g., medical imaging: tumour region segmentation improves diagnosis.

Why not simply crop the image manually?

Application: Better Classification

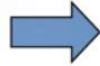


E.g., medical imaging: tumour region segmentation improves diagnosis.

Why not simply crop the image manually?

Manual cropping is inconsistent and time-consuming; segmentation automates this reliably.

Application: Speed up Recognition



[Felzenszwalb and Huttenlocher 2004]

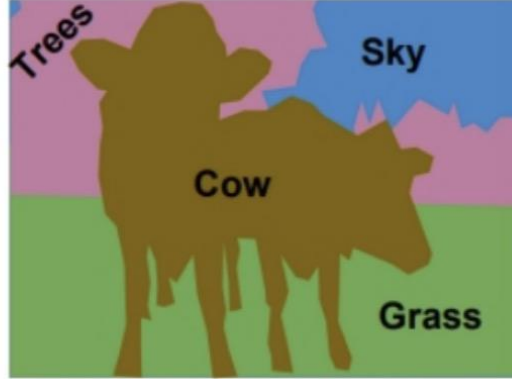
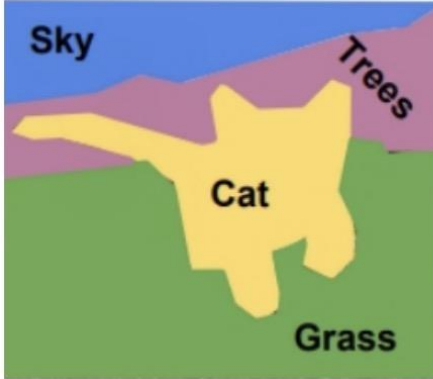


[Shi and Malik 2001]

[Hoiem et al. 2005, Mori 2005]

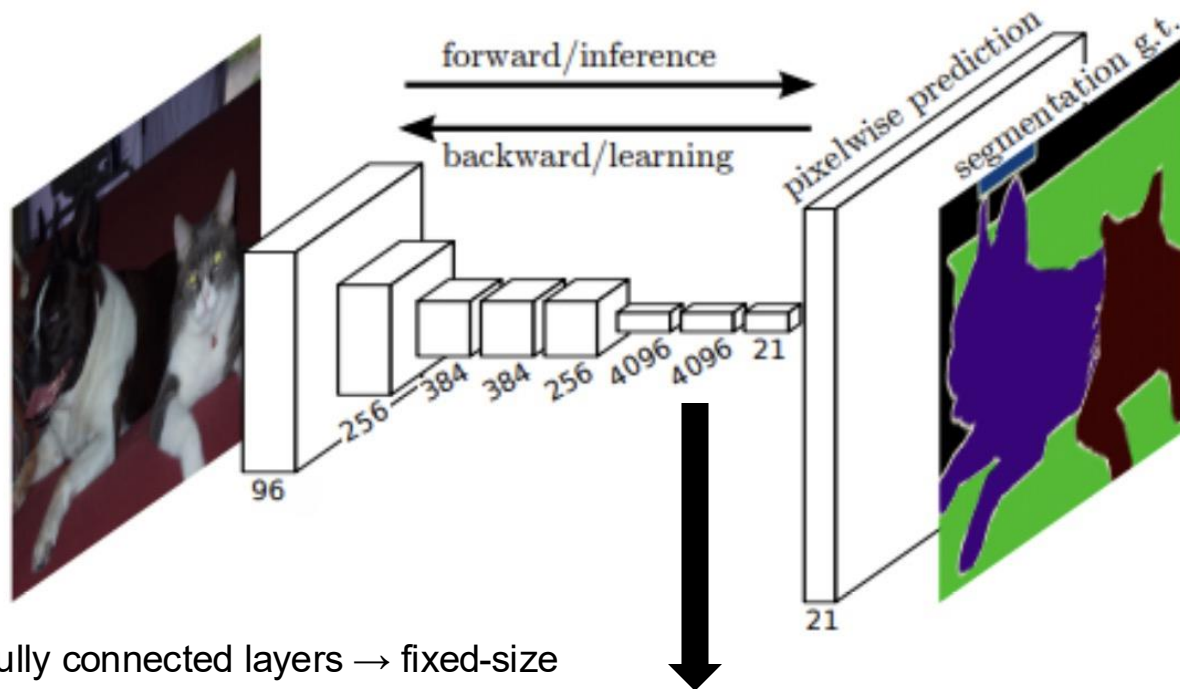
In autonomous driving, segmenting road/lane/sky reduces unnecessary detection attempts.

Semantic Segmentation



- Every pixel in the image needs to be labelled with a category label
- Do not differentiate between the instances (see how we do not differentiate between pixels coming from different cows)

Fully convolutional networks (FCN)

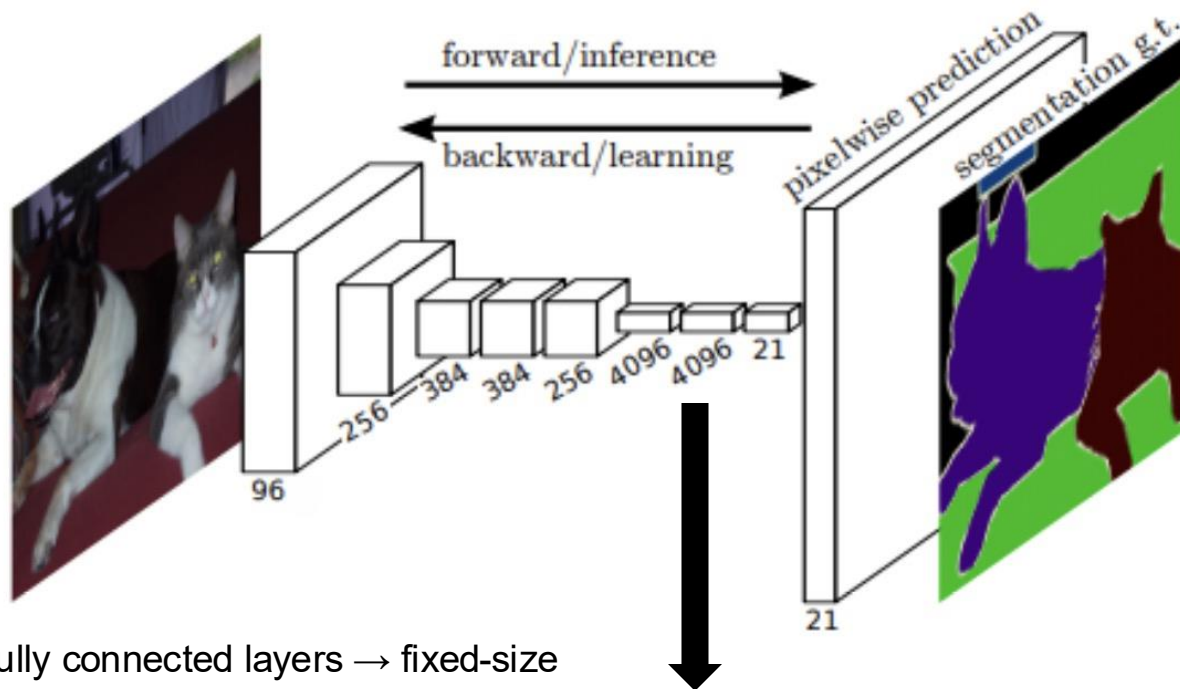


The original CNN had fully connected layers → fixed-size input, coarse output. FCN removes FC layers so the network becomes fully convolutional.

fully connected layers in CNN

Why do fully connected layers break pixel-level prediction?

Fully convolutional networks (FCN)



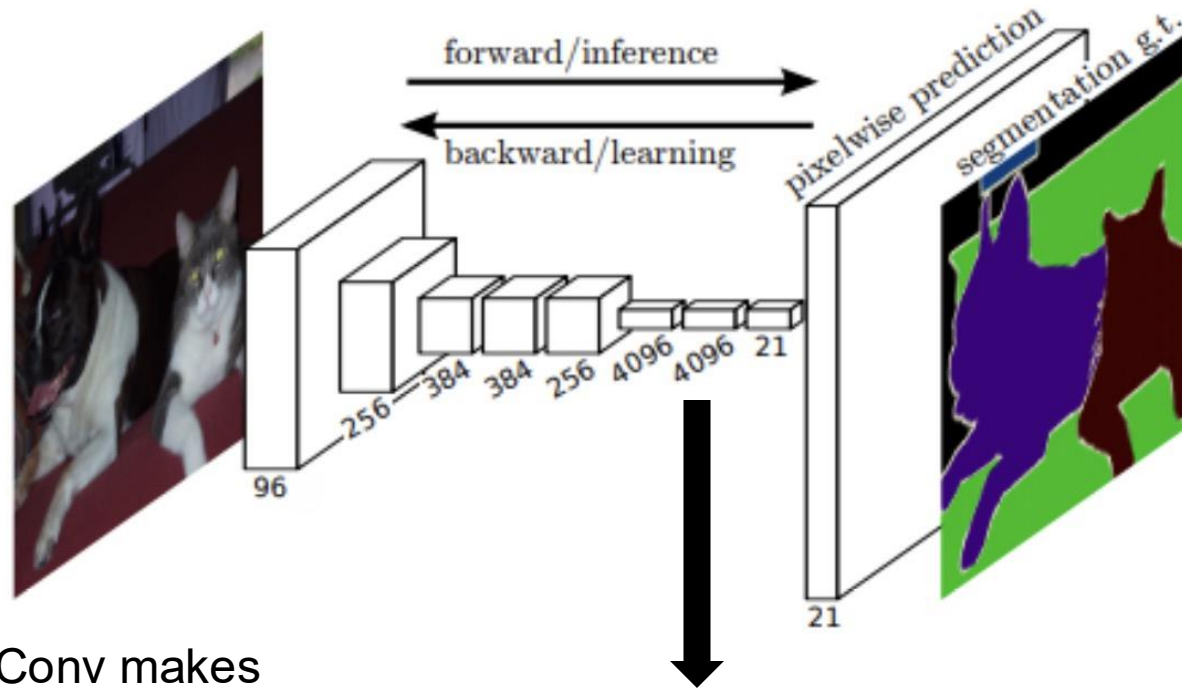
The original CNN had fully connected layers → fixed-size input, coarse output. FCN removes FC layers so the network becomes fully convolutional.

fully connected layers in CNN

Why do fully connected layers break pixel-level prediction?

They remove spatial information because they flatten feature maps.

Fully convolutional neural networks

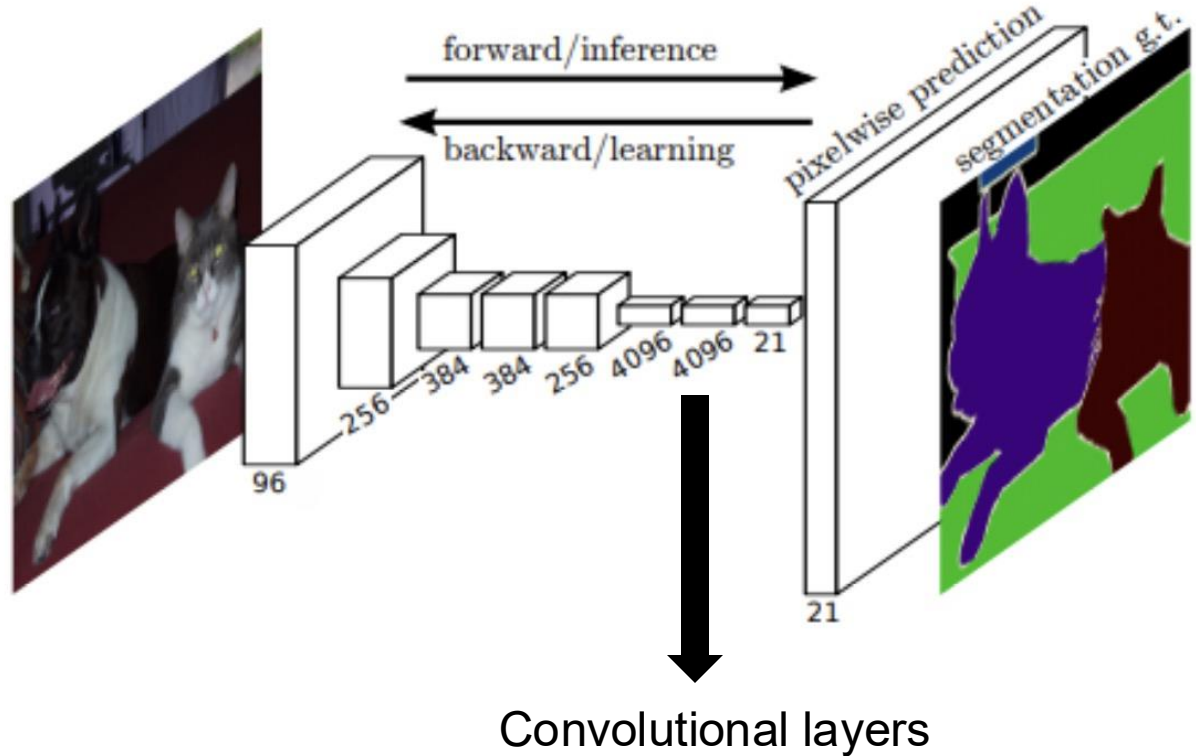


Converting FC→Conv makes segmentation possible on entire images.

Convolutional layers

Fully convolutional neural networks

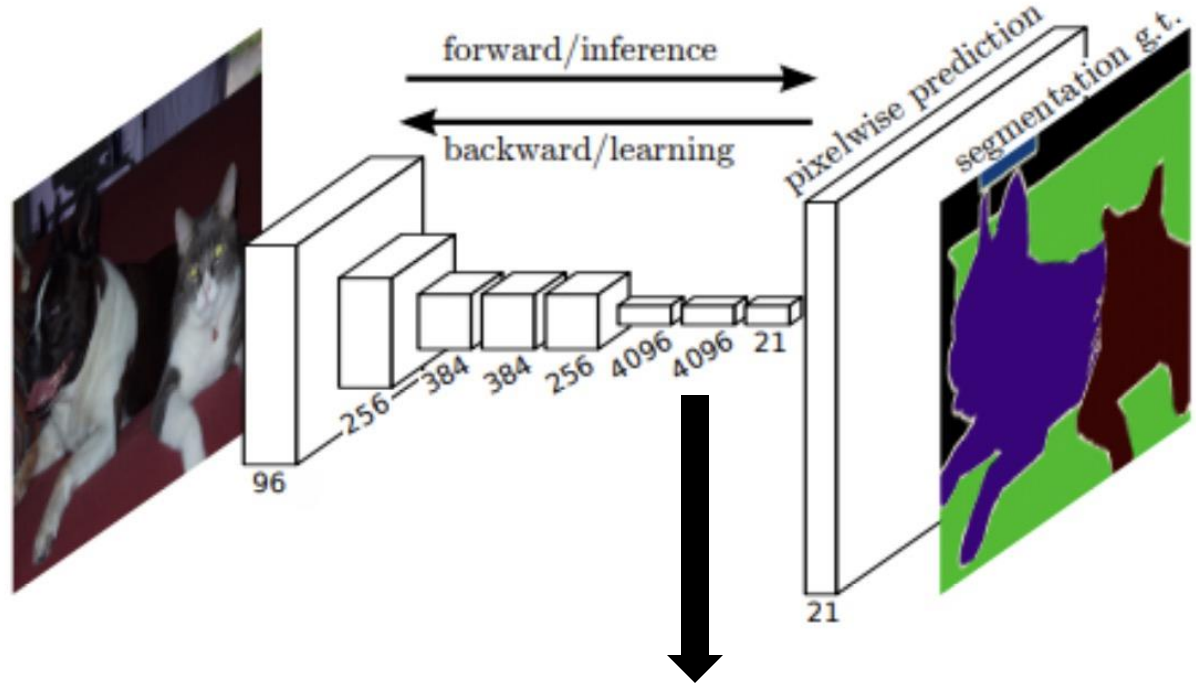
1. Replace FC layers with convolutional layers.
2. Convert the last layer output to the original resolution.
3. Do softmax-cross entropy between the pixelwise predictions and segmentation ground truth.
4. Backprop and SGD



What loss function is commonly used in segmentation?

Fully convolutional neural networks

1. Replace FC layers with convolutional layers.
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3. Do softmax-cross entropy between the pixelwise predictions and segmentation ground truth.
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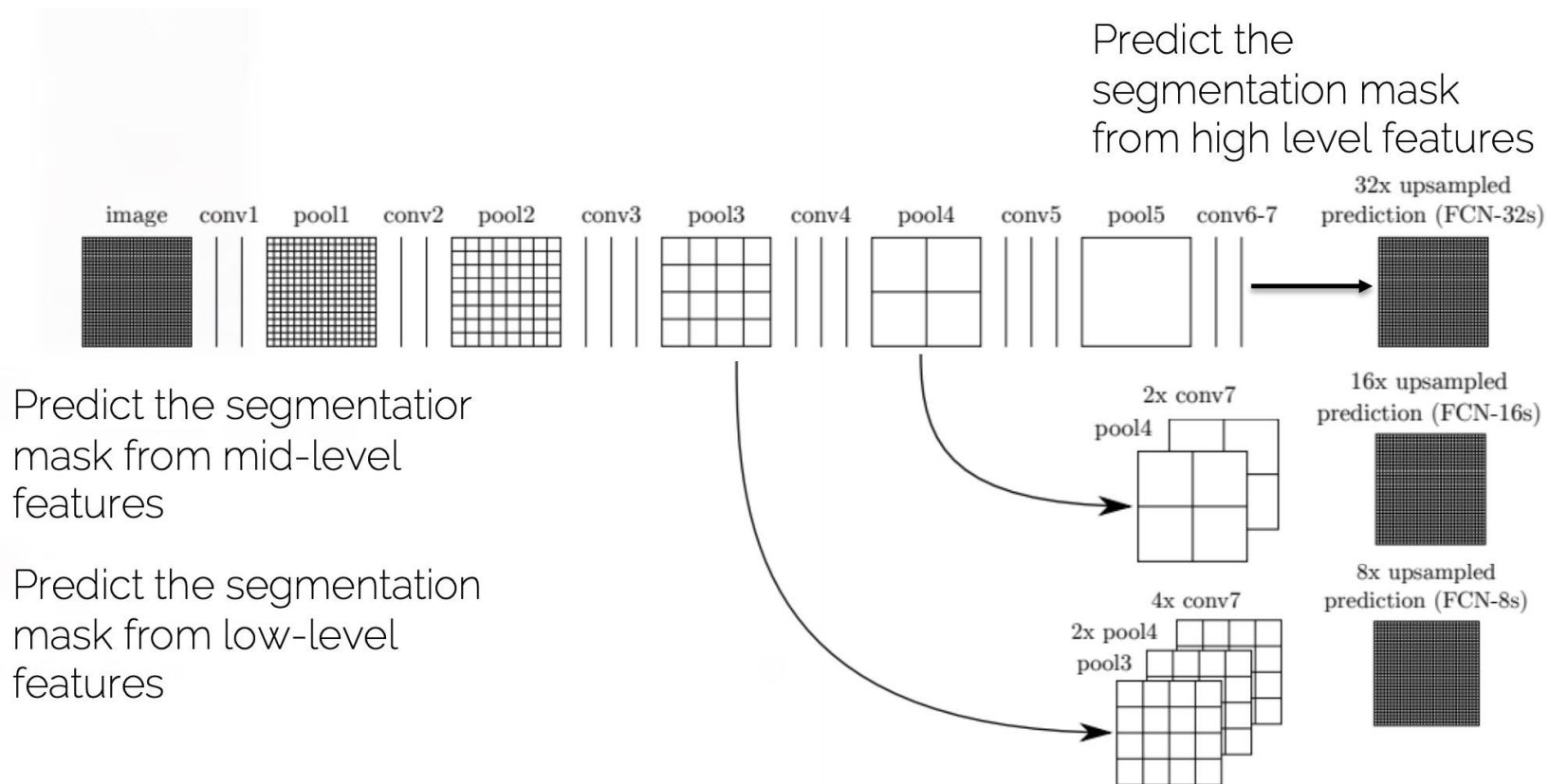


Convolutional layers

What loss function is commonly used in segmentation?

Pixelwise softmax-cross-entropy.

Network architecture



Neighbour Upsampling

Nearest Neighbor

1	2
3	4



1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Input: 2 x 2

Output: 4 x 4

Is nearest-neighbour
upsampling
learnable?

Neighbour Upsampling

Nearest Neighbor

1	2
3	4



1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Input: 2 x 2

Output: 4 x 4

Is nearest-neighbour
upsampling
learnable?

No — it uses a
fixed rule, not
parameters.

Unsampling-Transposed convolution

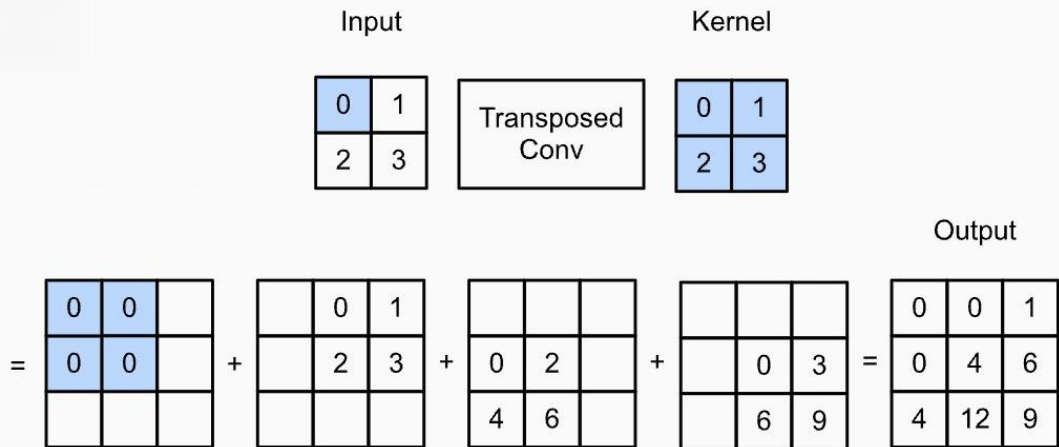


Fig. 14.10.1 Transposed convolution with a 2×2 kernel. The shaded portions are a portion of an intermediate tensor as well as the input and kernel tensor elements used for the computation.

Unsampling-Transposed convolution

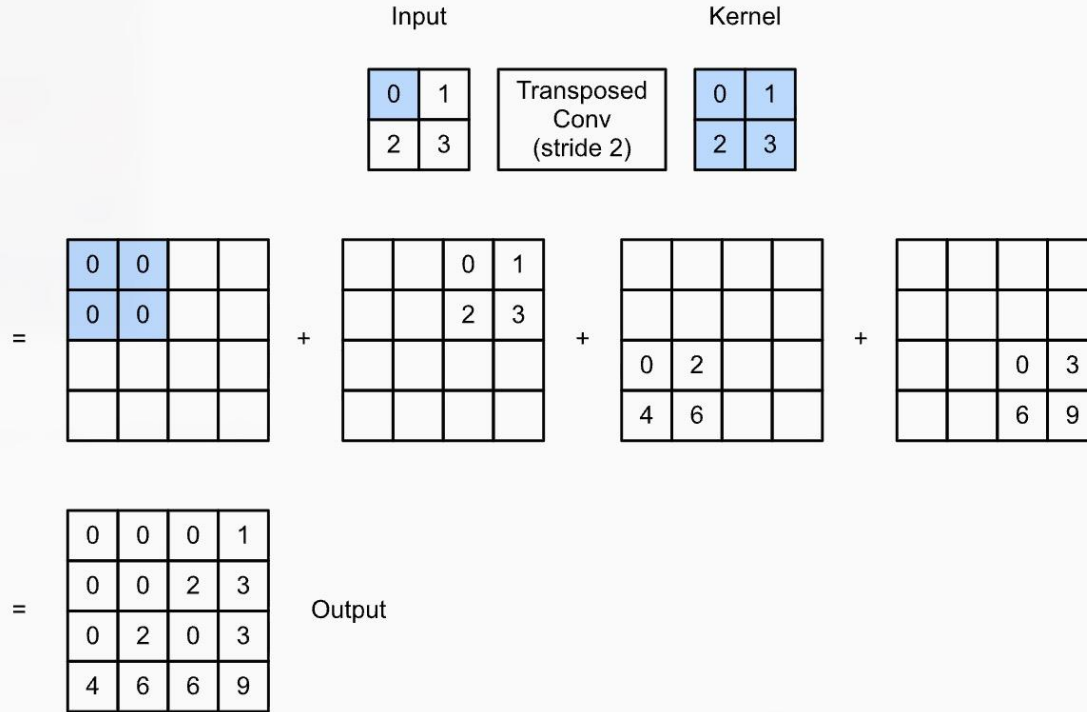
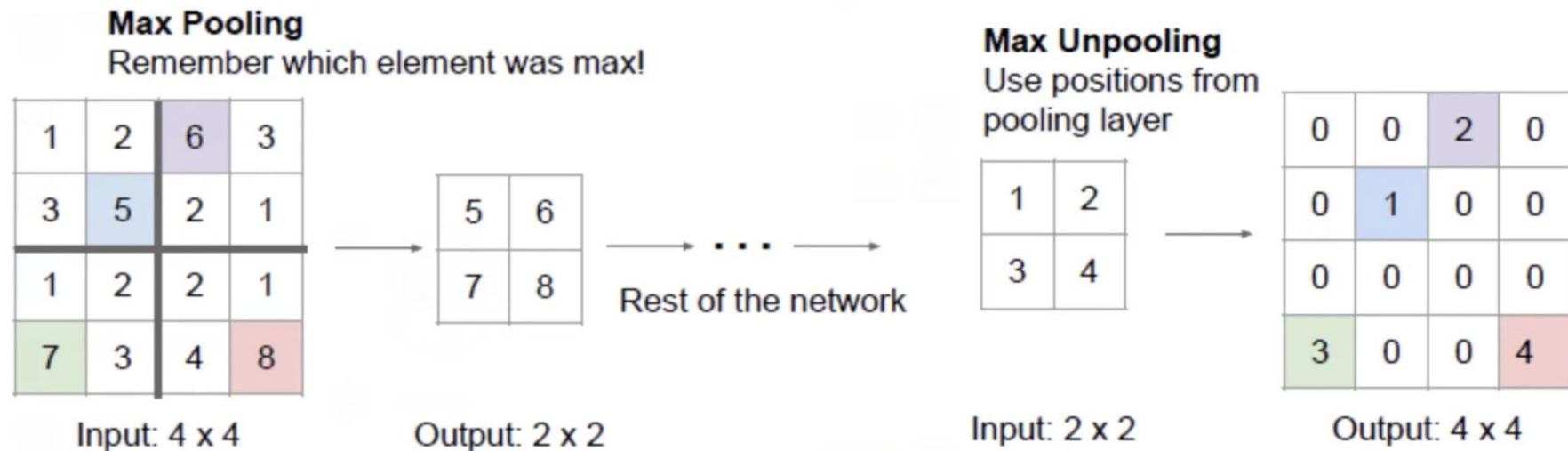


Fig. 14.10.2 Transposed convolution with a 2×2 kernel with stride of 2. The shaded portions are a portion of an intermediate tensor as well as the input and kernel tensor elements used for the computation.

Unsampling-Unpooling



Qualitative results

Fully Convolutional Network with a 32/16/8-stride unsampling

FCN-32s



Good

High-level features

FCN-16s



Better

Mid-level features

FCN-8s



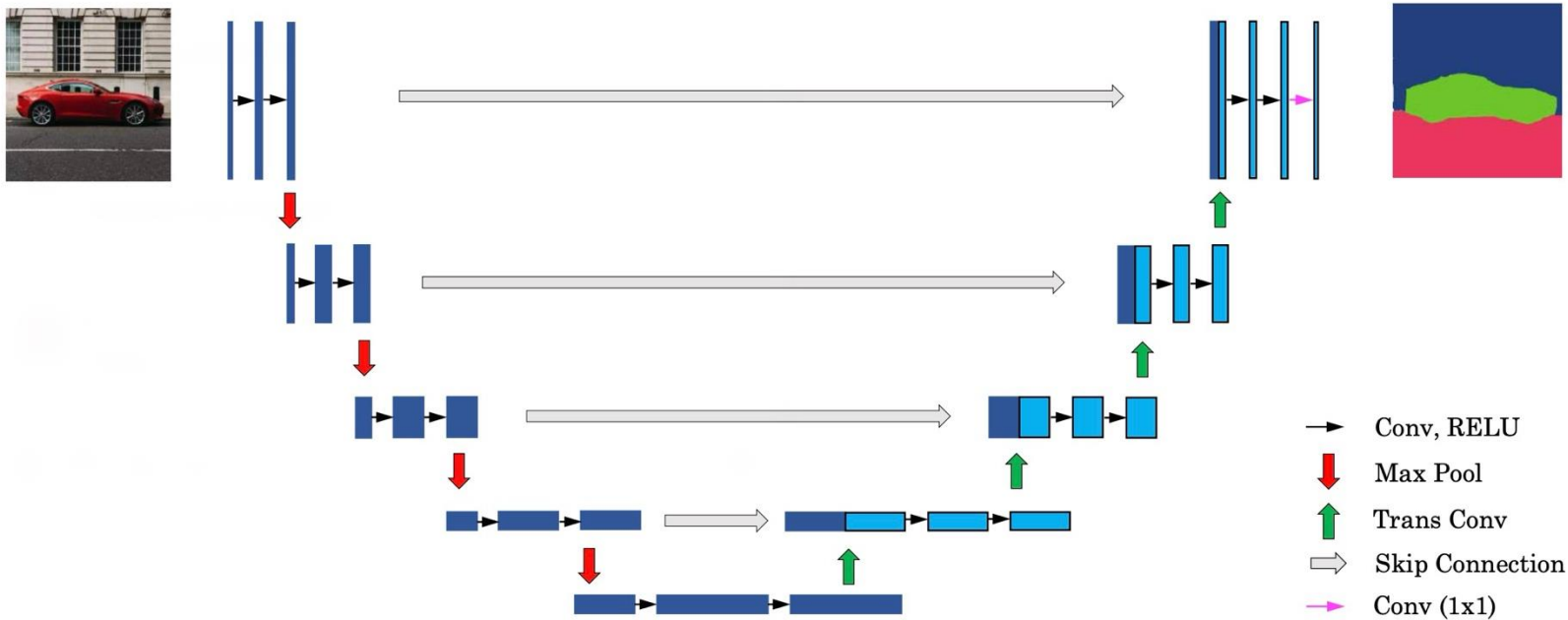
Best

Low-level features

Ground truth

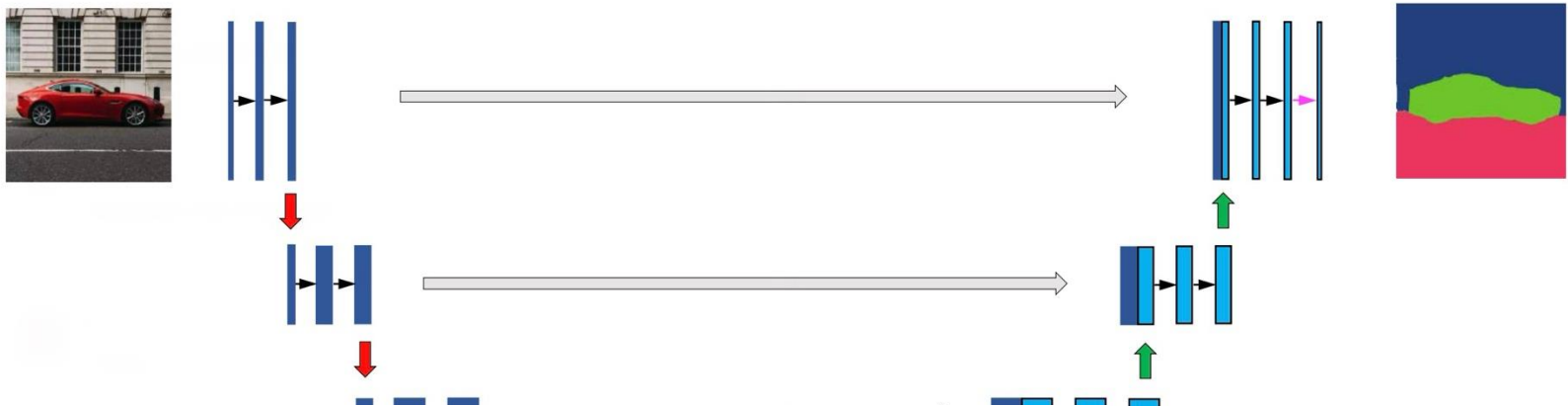


Unet

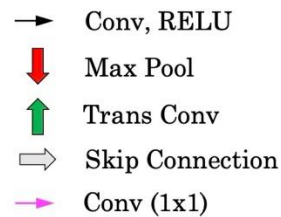


U-Net is extremely popular in medical imaging. It uses skip connections to combine low-level spatial detail with high-level semantic information.

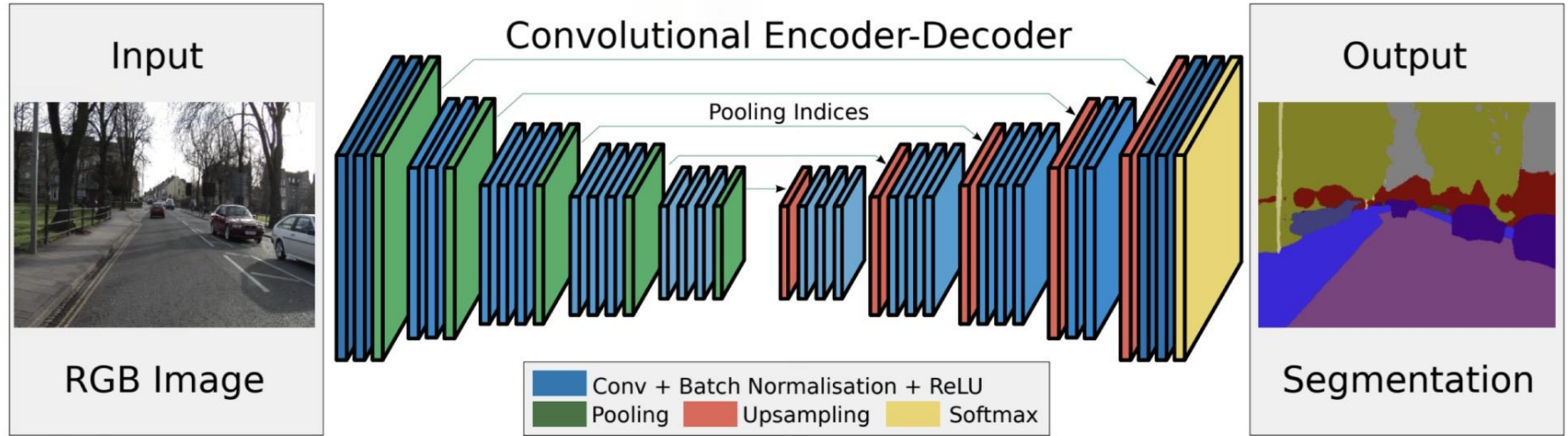
Unet



- Encoder reduces spatial resolution
- Decoder increases resolution
- Skip connections:
 - Combine low-level details + high-level semantics
 - Better boundary reconstruction



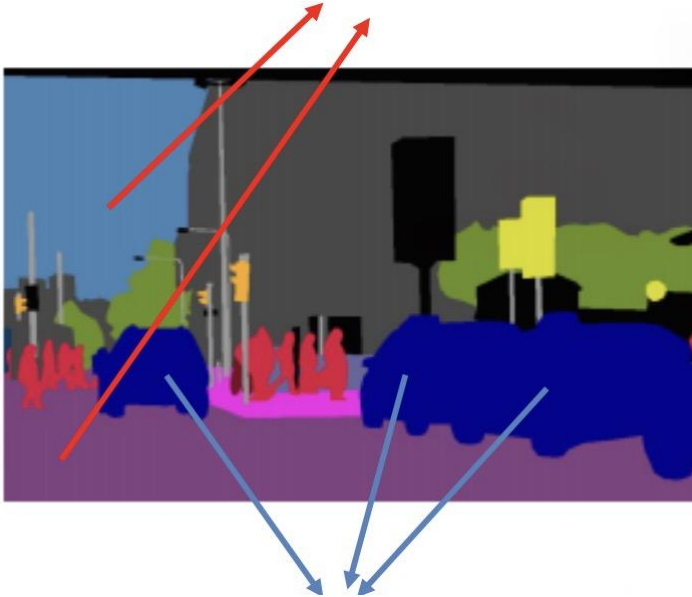
SegNet



SegNet stores max-pool indices to make unpooling precise. It also reduces memory compared to U-Net and works well in segmentation tasks where boundaries matter.

Instance Segmentation

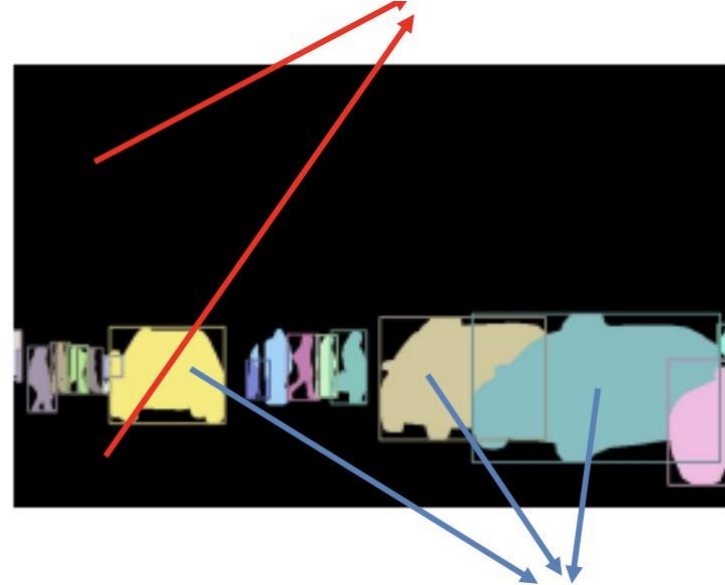
Label every pixel, including the background (sky, grass, road)



Do not differentiate between the pixels coming from instances of the same class

Semantic

Do not label pixels coming from uncountable objects (sky, grass, road)

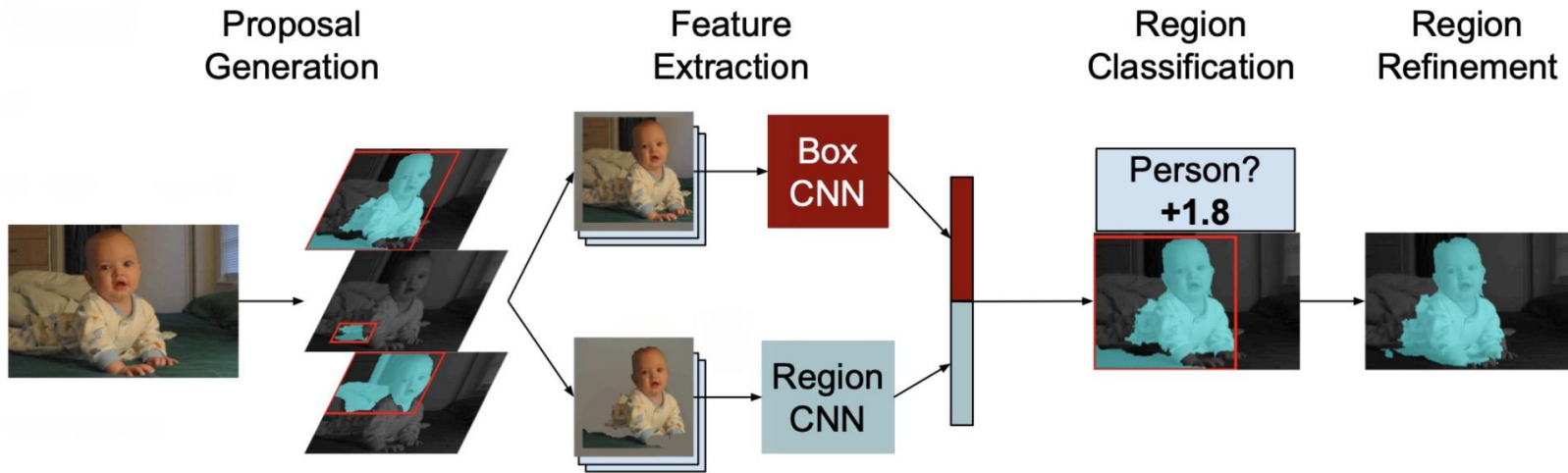


Differentiate between the pixels coming from instances of the same class

Instance

Instance Segmentation

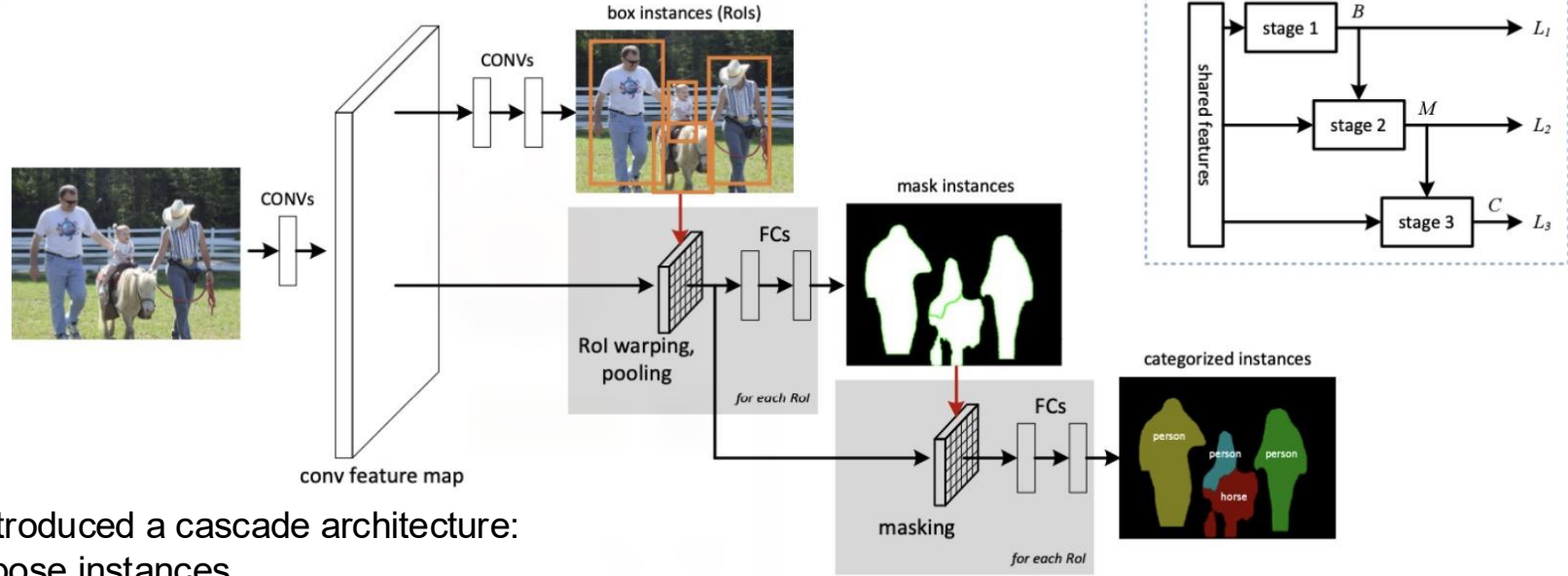
- SDS: Simultaneous Detection and Segmentation



SDS solves detection and segmentation jointly. This was one of the first successful pipelines.

Instance Segmentation

- MNC: Multi-task network cascades



MNC introduced a cascade architecture:

- Propose instances
- Refine masks
- Classify objects

Instance Segmentation

- YOLACT++

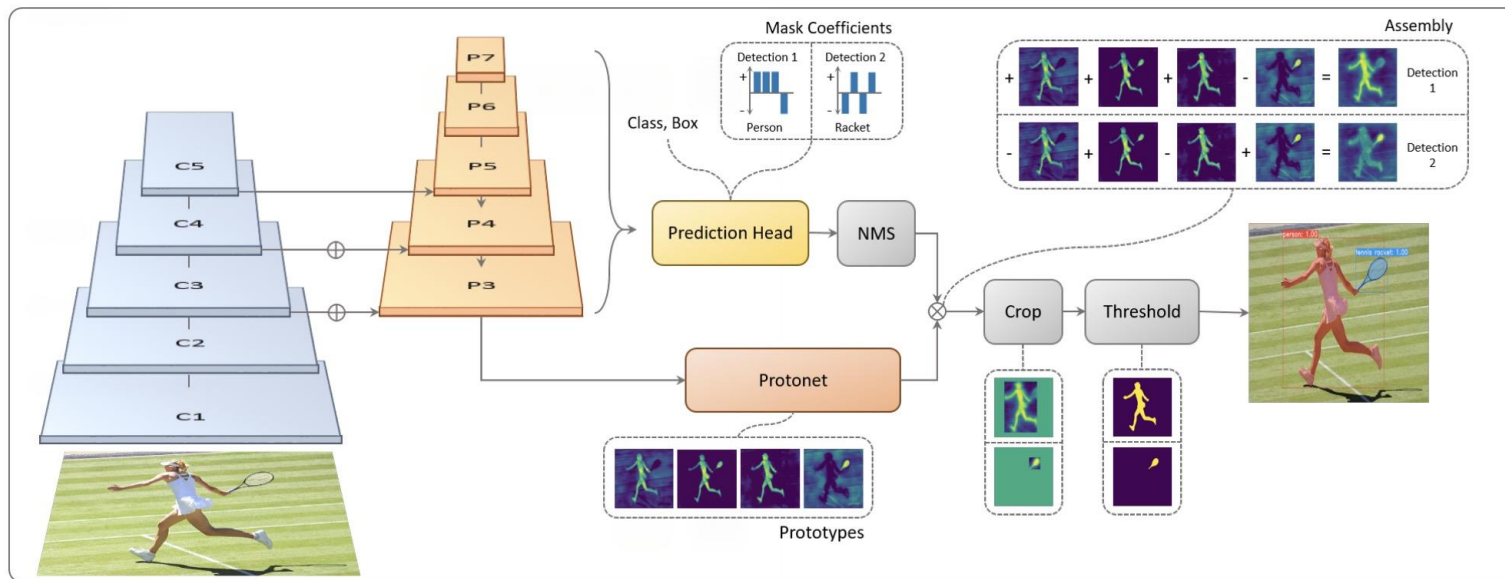
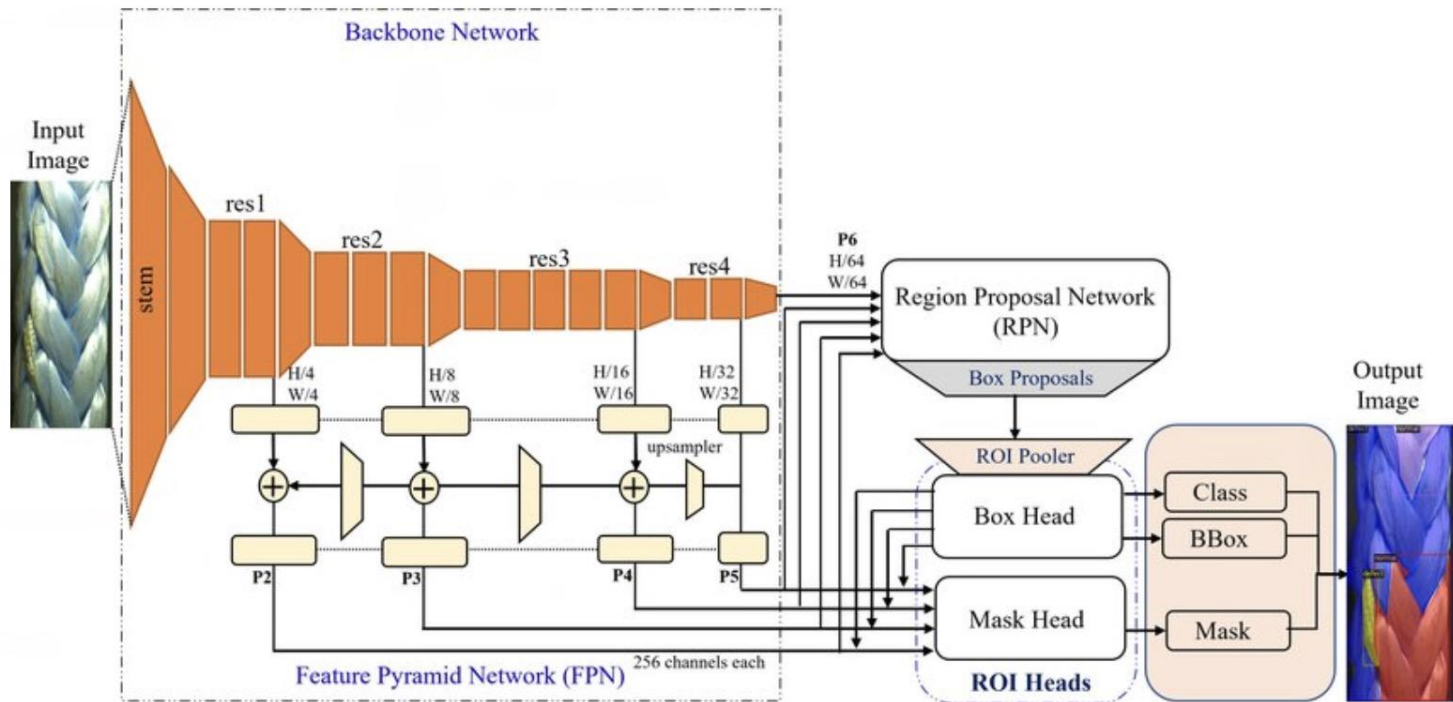


Fig. 2: **YOLACT Architecture** Blue/yellow indicates low/high values in the prototypes, gray nodes indicate functions that are not trained, and $k = 4$ in this example. We base this architecture off of RetinaNet [27] using ResNet-101 + FPN.

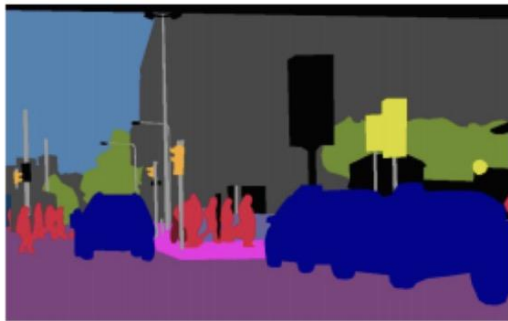
Instance Segmentation

- Detectron2



Panoptic Segmentation

Semantic
segmentation



+

Instance
segmentation



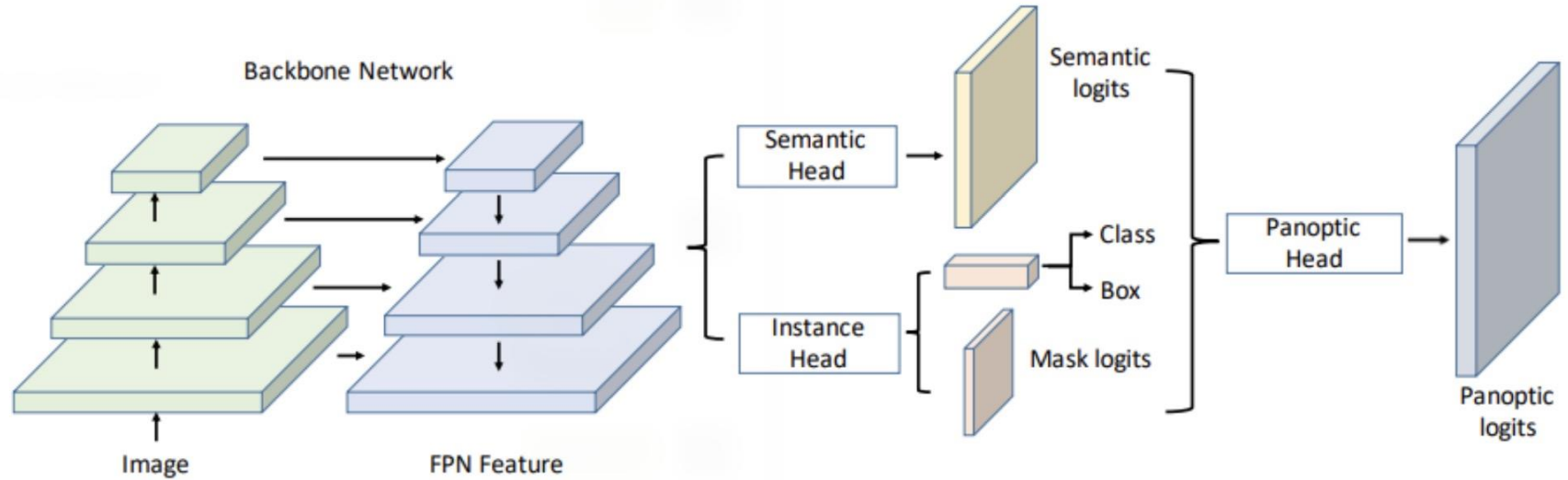
=

Panoptic
segmentation



Panoptic segmentation unifies semantic + instance segmentation. It tries to label everything *and* distinguish individual objects at the same time.

Panoptic Segmentation



Shared features

Separate heads

Putting it
together

Example Exam Questions

Question 2 — Optimisation & Backpropagation (10 marks) ←

Explain the vanishing gradient problem. ↓

Describe (i) why it occurs, and (ii) how to mitigate it. ←

Example Exam Questions

Question 2 — Optimisation & Backpropagation (10 marks)↵

Explain the vanishing gradient problem.↓

Describe (i) why it occurs, and (ii) how to mitigate it.↵

Occurs when gradients shrink during backpropagation through many layers.↵

Mitigation technique: ↵

Residual connections: allow gradients to flow directly by skip connections.↵

Example Exam Questions

Question 3 — Regularisation in Deep Learning (20 marks) ↵

Discuss the following regularisation techniques and explain when each is most appropriate: ↵

(a) Data augmentation ↵

(5 marks) ↵

(b) Early stopping ↵

(5 marks) ↵

(c) Label smoothing ↵

(5 marks) ↵

(d) Weight decay (L2 penalty) ↵

(5 marks) ↵

Example Exam Questions

Question 3 — Answers

(a) Data augmentation (5 marks)

Expands training set via flips, crops, noise.

Best for: vision/audio tasks where invariances exist.

(b) Early stopping (5 marks)

Stop training when validation error increases.

Best for: preventing overfitting when model learns too quickly.

(c) Label smoothing (5 marks)

Replace one-hot labels with softened targets (e.g., 0.9/0.1).

Best for: classification tasks prone to overconfidence.

(d) Weight decay (L2) (5 marks)

Penalises large weights → smoother functions.

Best for: general-purpose regularisation in most deep models.

Example Exam Questions

Question 6 — Convolution & Parameters (20 marks)↵

A convolutional layer receives a 7×7 input with 3 channels.↵

It applies 4 filters of size 3×3 , stride 2, and padding 1.↵

↵

(a) Compute the spatial output size.↵

(5 marks)↵

↵

(b) How many values does the final output volume contain?↵

(5 marks)↵

↵

(c) Compute the total number of trainable parameters (including biases).↵

(5 marks)↵

↵

(d) Explain why increasing stride reduces computation and list one disadvantage.↵

(5 marks)↵

Example Exam Questions

Question 6 — Answers

Given

Input: 7×7

Filters: 3×3

Stride = 2

Padding = 1

(a) Output spatial size

Formula:

$$\frac{W - K + 2P}{S} + 1 = \frac{7 - 3 + 2}{2} + 1 = 4$$

Output = 4×4

(b) Total outputs

4 filters $\rightarrow$ 4 channels

Each channel: $4 \times 4 = 16$

Total = **64 values**

(c) Trainable parameters

Each filter:

$3 \times 3 \times 3$ channels = 27 weights + 1 bias = 28 parameters

4 filters $\rightarrow 4 \times 28 = 112$

Total parameters = **112**

(d)

Increasing stride:

- reduces computation and memory (fewer positions computed)
- but disadvantages: reduces spatial detail, may harm accuracy

**I will put an example exam paper on ELE in
the next week.**

**Please leave your feedback to improve the
module.**

